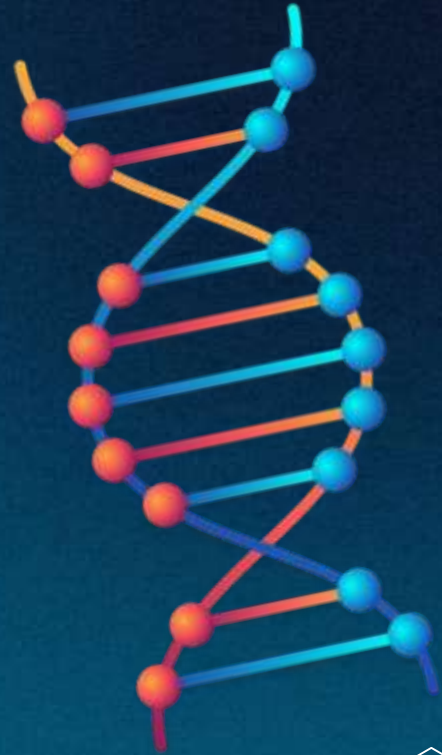




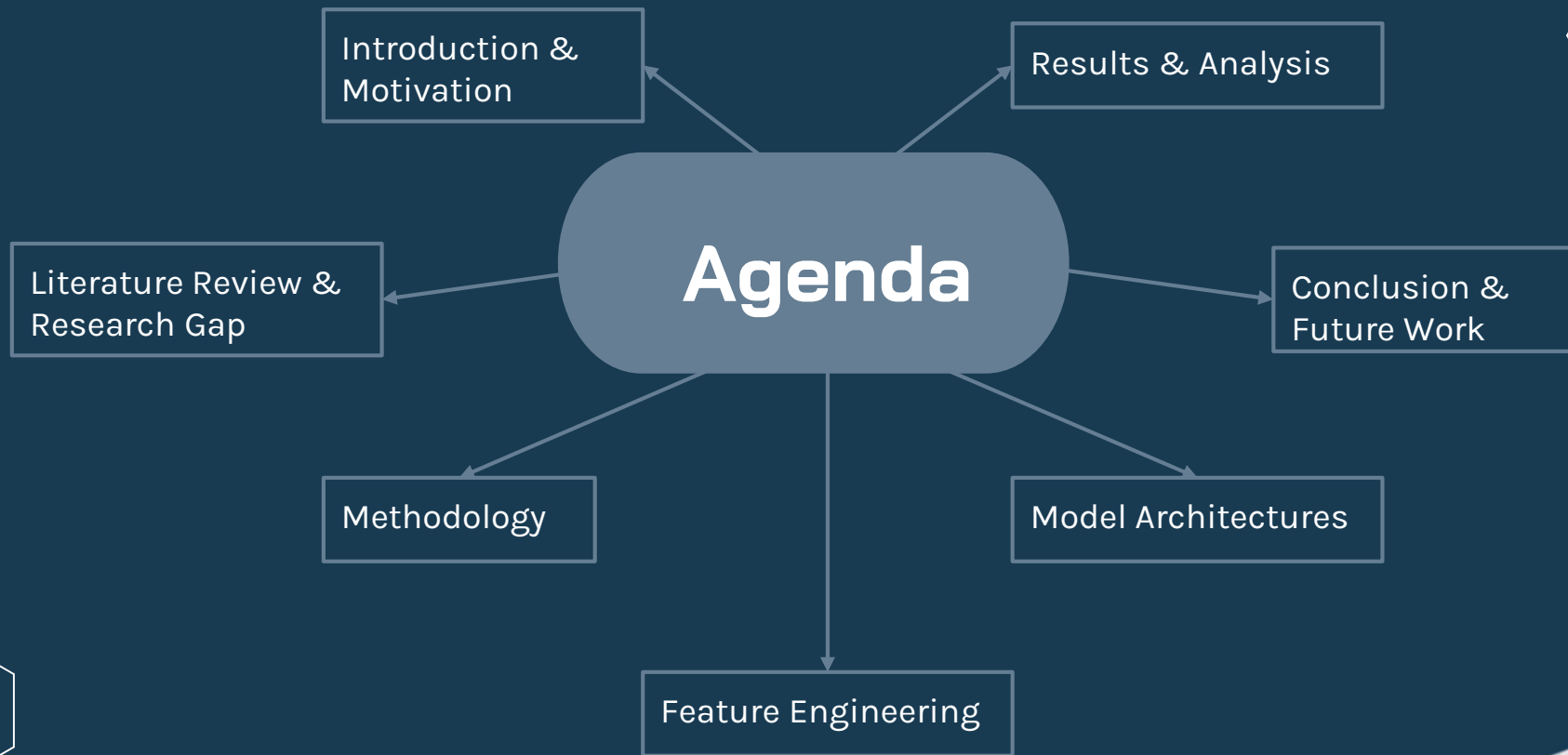
A Comparative Analysis of Machine Learning Architectures for **siRNA** **Efficacy Prediction**



Animesh Kumar (22CS30009)

Professor Pralay Mitra

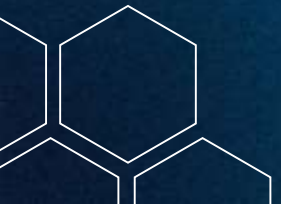
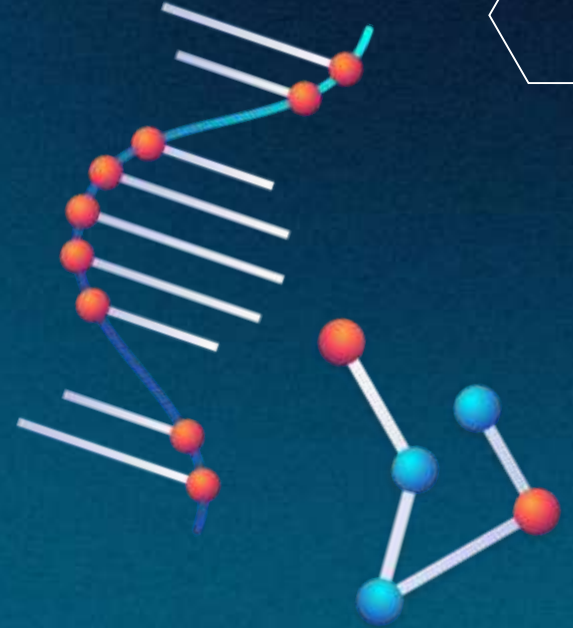


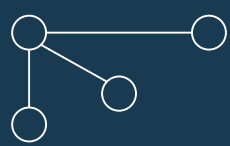




The Big Picture

The Promise of RNA
Interference





RNA Interference as a Therapeutic Tool

1. A diseased cell with an overactive gene.
2. An siRNA molecule being introduced.
3. The target mRNA being silenced.
4. A healthy cell.



A powerful, natural mechanism to selectively 'turn off' any gene of interest.



The Central Challenge: A Massive Search Problem

- For one gene, thousands of potential siRNA candidates exist.
- Efficacy varies dramatically (from >90% to <10%).
- Wet-lab screening is prohibitively expensive and time-consuming.

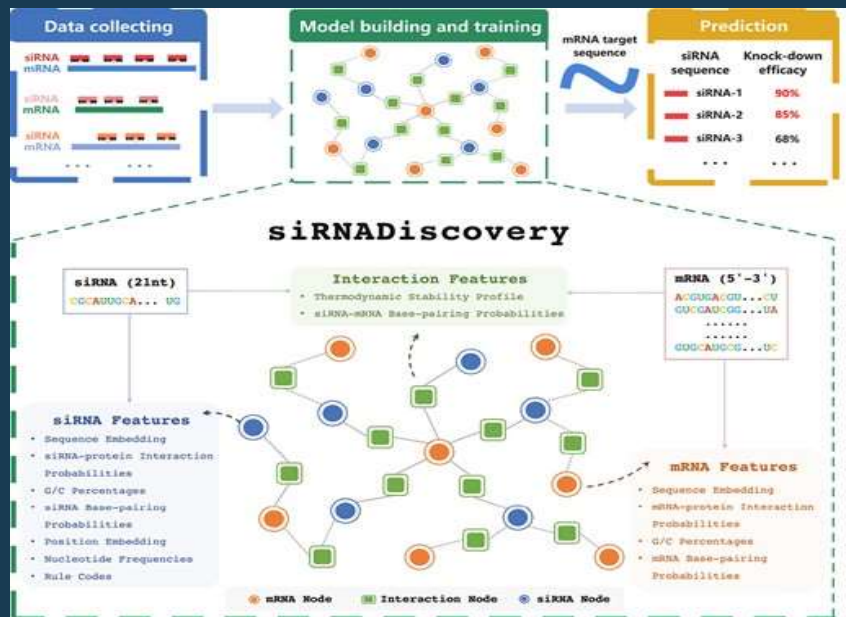
Accurate computational prediction is a vital necessity.



The State-of-the-Art: Modeling Biology as a Graph

The current state-of-the-art, siRNADiscovery, uses a Graph Neural Network (GNN) to learn from the relationships between these biological entities.

The GNN creates "context-aware" predictions by passing information between the nodes.



The Research Gap & Our Hypothesis

Two competing factors:

- ❖ **Rich Features:**

- The GNN's success relies on a massive, biologically-informed feature set (~39,000 features).

- ❖ **Complex Architecture:**

- The GNN itself is complex, slow to train, and acts as a "black box."

Question

Is the GNN's complexity necessary, or is the feature set the true driver of performance?

Hypothesis

We hypothesize that established, interpretable machine learning models can achieve competitive performance when given the same high-quality feature set.

Our Experimental Approach: A Comparative Analysis

1. Ensemble Models:

- XGBoost
- LightGBM
- Random Forest

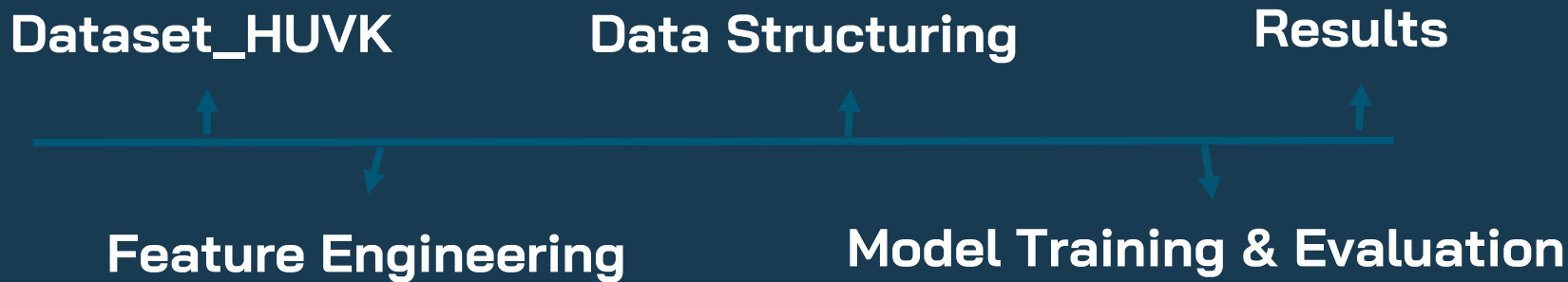
1. Hybrid Graph-based Model:

- Spektral (GNN)





A High-Level Overview



The same feature set was used for all models; only the data structure and model architecture were changed.





Feature Engineering: The Foundation of Prediction

Four quadrants, each with a list of features:

1. **Sequence Features:** One-Hot Encoding, Positional Embedding
2. **Structural Features:** Base-Pairing Probabilities (Self-fold, Co-fold)
3. **Biophysical Features:** Thermodynamic Stability, RNA-AGO2 Interaction
4. **Empirical Rules:** k-mer Frequencies, G/C Content, Rule Codes





Data Structuring for Different

A **For Ensemble Models (XGBoost, LGBM, RF)**

Flattened Feature Matrix

1. Feature Aggregation: All features are collected:
 - siRNA-specific features
 - mRNA-specific features
 - Interaction-specific features
2. Concatenation: For each siRNA-mRNA pair, these features are combined into a single, long feature vector.
3. Result: A single, large tabular matrix where each row is an experiment and each column is a feature.
 - Dimensions: (Number of Samples) x (~39,000 Features)

For the Spektral (GNN) Model

Hybrid Graph-Tabular Method

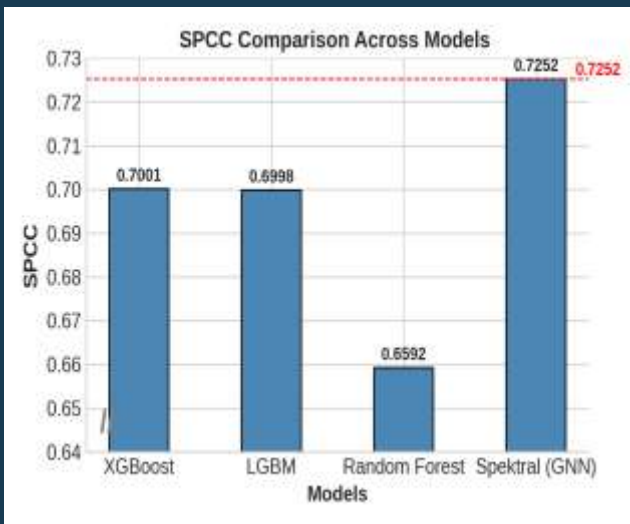
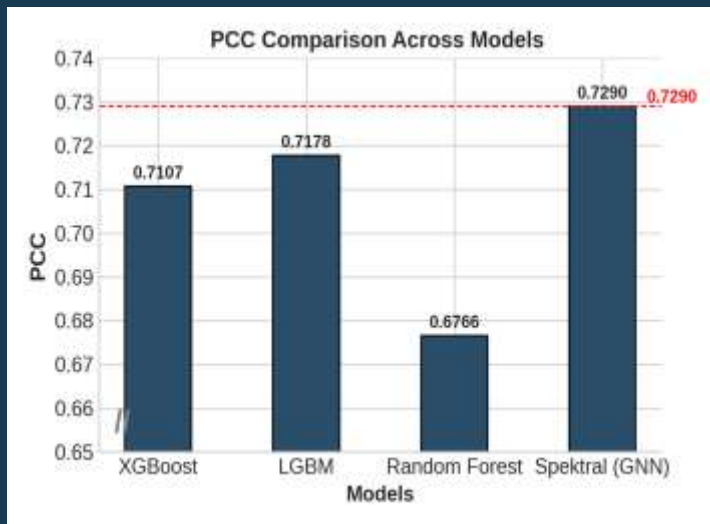
1. Graph Construction: First, a heterogeneous graph is built to represent the relationships between all siRNA, mRNA, and Interaction nodes.
2. Spektral embeddings are generated to convert the graph's structure into new features.
3. Result: An augmented tabular matrix that is fed into an XGBoost model.

Results: Overall Performance

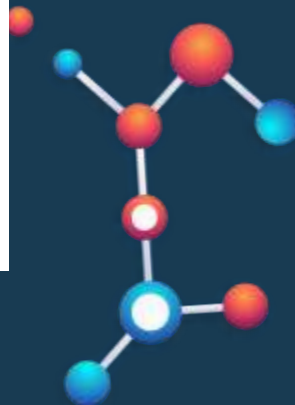
Model	PCC ↑	SPCC ↑	MSE ↓	AUC ↑
Spektral (GNN)	0.7290	0.7252	0.0228	0.8480
LightGBM	0.7178	0.6998	0.0221	0.8314
XGBoost	0.7107	0.7001	0.0225	0.8364
Random Forest	0.6766	0.6592	0.0249	0.8108

The Spektral (GNN) model is the best-performing among our implementations, with LightGBM as a close second.

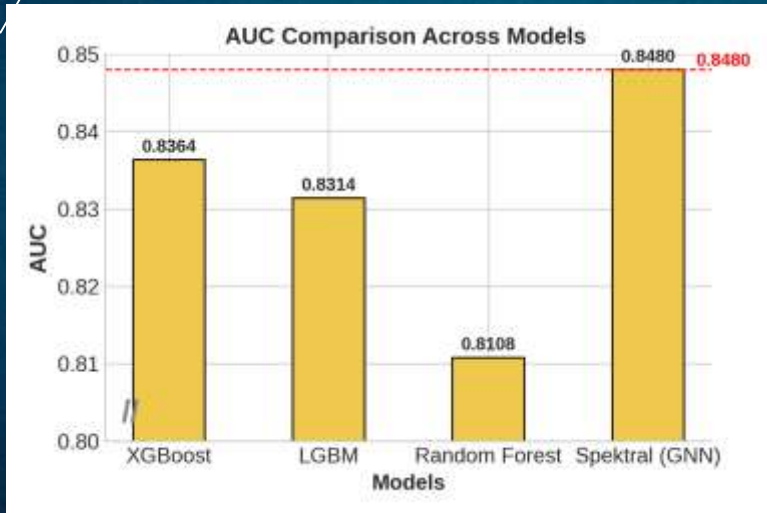
Analysis by Metric: Correlation



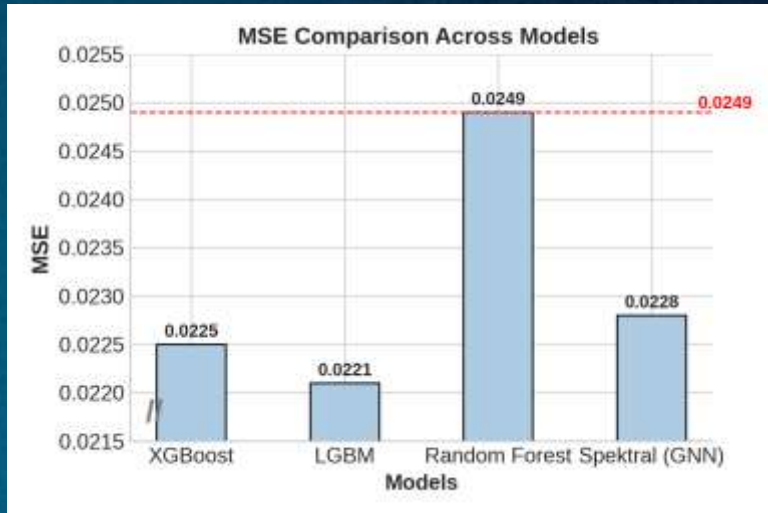
The Spektral (GNN) model is the clear winner, demonstrating that encoding the graph's topology as features significantly improves the model's ability to correctly rank siRNAs.



Analysis by Metric: Prediction Error (MSE) & Classification (AUC)



The results are much closer on this metric. LightGBM achieves the lowest error (best performance).



The Spectral (GNN) model is again the top performer. Standard XGBoost and LightGBM are also strong classifiers.



Key Findings & Analysis

Graph Structure Boosts Performance

Observation:

Adding graph-derived spektral features to the XGBoost model significantly improved its predictive power.

Result:

The Pearson Correlation Coefficient (PCC) increased from 0.711 (standard XGBoost) to 0.729 (Spectral-GNN).

Conclusion:

This directly demonstrates that the underlying network topology contains unique and valuable predictive information that is not captured by the node features alone.

Ensemble Models are Highly Competitive & Practical

Observation:

Standard gradient boosting models, without any graph information, performed exceptionally well.

Result:

LightGBM achieved a PCC of 0.718, very close to the more complex Spectral model's 0.729, and delivered the lowest overall MSE.

Conclusion:

For practical applications, a well-tuned ensemble model on a flattened feature set offers a powerful and efficient solution with less implementation overhead.

The Advantage of Interpretability

Key Insight:

Unlike "black-box" deep learning models, ensemble methods like XGBoost and Random Forest can rank the importance of every feature.

Significance:

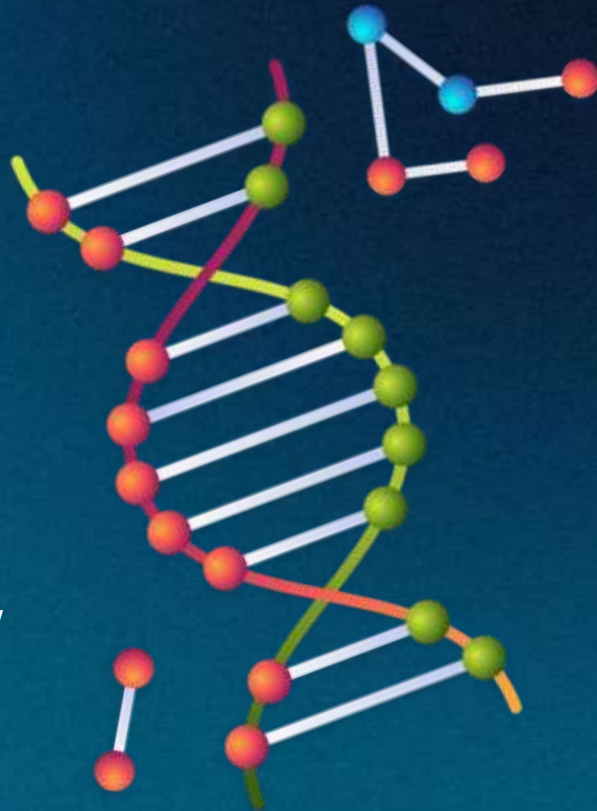
This allows us to identify the specific biological factors that are most predictive of siRNA efficacy.

Conclusion:

This interpretability provides direct, actionable insights for scientific discovery, which is a crucial advantage for any model used in a biological context.

Conclusion

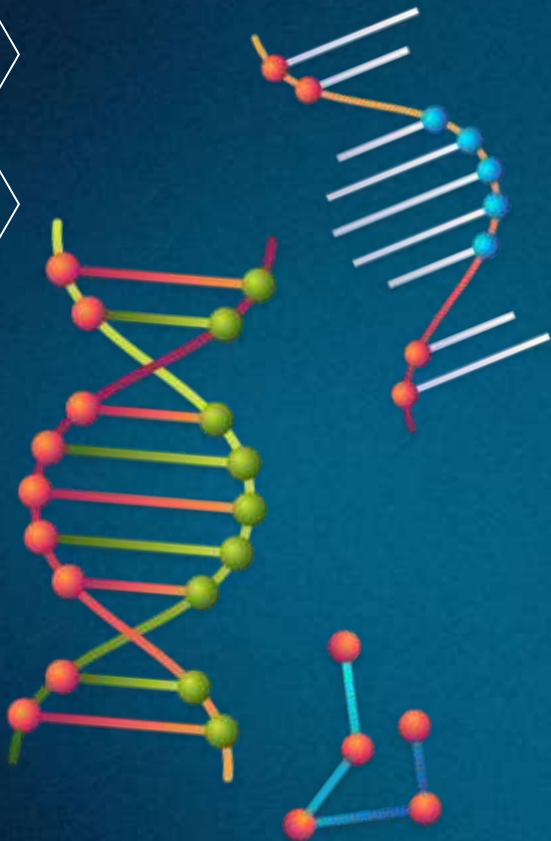
- This project successfully compared four distinct ML architectures for **siRNA efficacy** prediction.
- The hybrid **Spektral (GNN)** model achieved the highest correlation, demonstrating the value of graph structural information.
- Standard gradient boosting models (**LightGBM/XGBoost**) are highly competitive, offering a powerful and practical alternative with less complexity.
- Ensemble models provide crucial interpretability through feature importance, a key advantage for scientific discovery.





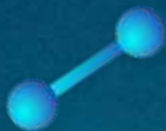
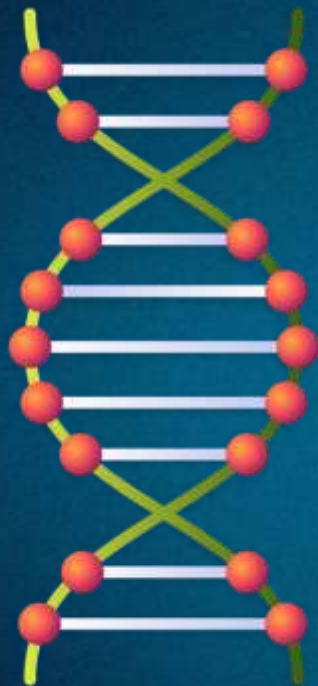
Future Work

- **Advanced Hyperparameter Optimization:** Use frameworks like Optuna to systematically tune all models and potentially close the performance gap with the SOTA baseline.
 - **Deeper Investigation into GNNs:** Explore more advanced GNN architectures like Graph Attention Networks (GATs).
 - **Feature Selection:** Use the feature importance rankings from our models to conduct a feature selection study to create smaller, more efficient models.
 - **External Validation:** Test the generalizability of all models on the external Simone and in-house (ACON) datasets.
- 



Thank You





Question

Answer

